

Software Engineering

Instructor

Shamaila Qayum

Contact Detail

sqayyum@numl.edu.pk

Office Location

Ghazali Block,
Ground Floor, Room 1

Course Overview

Classes	BSSE
Session	Fall 2022
Course Time	Monday 15:15 – 17:50
Credit Hours	03 (3,0)
Total Weeks	16
Prerequisite	none

Office Hours

08:30 to 9:20

Thursday

10:40 - 11:55

Monday

Course Description

In this course, students will learn about some of the most basic topics on software engineering. This course would cover the basic and agile software process models. It further goes into the details of different phases of these models as Requirements Engineering, Analysis, Design and Testing.

Course Learning Outcomes

At the end of this course, Students should be able to

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO1	Explain the software engineering concepts, principles, and process flow of the SDLC	Cognitive	2	2	A1, Q1, Midterm
CLO2	Explain various software development processes/methodologies	Cognitive	2	2	A2, Q2 Midterm, Final Term
CLO3	Analyze software engineering key areas	Cognitive	4	3	A3, Q3, Presentation/Project, Final Term

Course Schedule

Week		Topic	Material
Week 1	12 th Sep	Introduction to Software Engineering	<u>Lecture 1 - Introduction.pptx</u>
Week 2	19 th Sep	Software process models(Presecriptive and Agile)	Lecture 2 ,3- Prescriptive Models.ppt Lecture 4,5 - Agile models.pptx Lecture 6- Kanban Lean
Week 3	26 th Sep	Requirements Engineering	Lecture 7 - RequirementsEngineering.pptx Lecture 8- RE Process Use Cases.pdf
Week 4	3 rd Oct	Use Cases	Lecture 8- REProcess Use Cases.pdf Lecture 9-Use case POS Example.pdf usecases.pdf
Week 5	10 th Oct	Use Cases	lecture 10- FullDressUseCase.pdf Full Dress Use Case Template
Week 6	17 th Oct	OOAD: Business model Domain model Activity Diagram DFD	Lecture 11-OOAD.pdf Lecture 12 - Domain model.pdf lecture 13-Activitiy Diagram.pptx data flow diagram.pdf
Week 7	24 th Oct	OOAD: Sequence diagram System sequence diagram	Lecture 14 - SystemSequenceDiagram.pdf Lecture 15- SequenceDiagram.pdf

Week		Topic	Material
Week 8	31st Oct	Class Diagrams/CRS Design concepts and patterns	Lecture 16-Class Diagram.pdf Lecture 17- CRC.pdf
Week 9		-Mid term exam-	
Week 10	21 st Nov	Testing: Testing tactics Testing techniques White box testing Black box testing	Lecture18- Testing Strategy.ppt Lecture 19-testing techniques.ppt
Week 11	28 th Nov	Project management: Risk management Managing people Team work	Lecture 20 - Project managment.pdf
Week 12	5 th Dec	Project planning: Software pricing Plan driven development Project scheduling Agile planning Estimation technique	Lecture 21 - Project Scheduling.ppt
Week 13	12 th Dec	Risk management Quality management: Software quality Software standards	Lecture 22 - Risk Management.ppt Lecture 23 - Quality management.pptx
Week 14	19 th Dec	-Presentation-	Presentation topics.docx Presentationschedule.docx
Week 15	26 th Dec	-Presentation-	Presentation topics.docx Presentationschedule.docx
Week 16		-Revision-	

Course Report/Degree Project

Students are required to prepare and present one of the advanced topics in SE. The timeline for the presentations is as follows

Deliverables	Due Date
Topic	<u>26th September 2022</u>
1 st Draft submission	<u>1st week of November 2022</u>
In-class oral presentations	<u>Mid Dec 2022 onwards</u>

Recommended Textbook

SOMMERVILLE, IAN. "SOFTWARE ENGINEERING 10TH EDITION."

To get ebook, you may download it from:

<http://dinus.ac.id/repository/docs/ajar/Sommerville-Software-Engineering-10ed.pdf>

Reference Material

SOFTWARE ENGINEERING; A PRACTITIONER'S APPROACH, BY ROGER. S. PRESSMAN, 6TH EDITION

To get ebook, you may download it from:

<https://ia600801.us.archive.org/23/items/SoftwareEngineeringAPractitionersApproach6thEditionRogerS.Pressman2/Software%20Engineering%20A%20Practitioner%27s%20Approach%206th%20edition%20Roger%20S.%20Pressman%20%282%29.pdf>

Teaching Methodology

- Lecturing, case studies.
- Activities based on case studies.

Grading Policy

Your final grade will depend on the following:

- Attendance and participation in class discussions
- Scores in assignment and quiz.
- The quality of your oral presentation.

However, following calculations are necessary for the final evaluation:

- Mid Term Exam: 25%
- End Term Exam: 40%
- Assignment and Quiz: 20%
- Presentation: 15%

Academic Honesty and Cheating

The University, the Faculty, and the teaching staff take cheating, plagiarism and other forms of academic fraud very seriously.

- Note 1: No late academic deliverables will be accepted.
- Note 2: There are no "make up" tests. Tests missed for no reason are deemed to have been written and failed and are marked "F".